Reg. No. | | | | | | | | | | |

[18AS41]

FOURTH SEMESTER B.Tech EXAMINATION, JULY – 2022

THERMAL SCIENCE

Time: 3 Hours **Maximum Marks: 70**

Instructions:
i. Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 14 marks.
ii. Missing data may be suitably assumed.

1. a. Differentiate intensive and extensive property with few examples. **06**

b. Freshwater and sea water are flowing in parallel horizontal pipes which are connected to each other by a double U tube manometer. Determine the pressure difference between the two pipelines. Take the density of sea water at that location to be ρ=1035 kg/m^3. Can the air column be ignored in the analysis? **08**

2. a. Consider a 1400 kg car cruising at constant speed of 70 km/hr. Now the car starts to pass another car by accelerating to 110 km/hr in 5 sec. Determine the additional power needed to achieve this acceleration. What would your answer be if the total mass of the car were only 700 kg. **06**

b. A piston cylinder device initially contains 0.07 m^3 of nitrogen gas at 130 kPa and 120°C. The Nitrogen is now expanded polytropically to a state of 100 kPa and 100°C. Determine the boundary work done during the process. Assume: $R_{Nitrogen}$=0.2986 kJ/kgK. **08**

3. a. Differentiate between open and closed system. **05**

b. Carbon dioxide enters an adiabatic nozzle steadily at 1MPa and 500°C with a mass flow rate of 6000 kg/hr and leaves at 100 kPa and 450 m/s. The inlet area of the nozzle is 40 cm^2. Determine: **07**
i. the inlet velocity
ii. the exit temperature.

c. What is the principle of entropy? Mention its units. **02**

OR

4. a. A 30 kg iron block initially at 450°C is quenched in an insulated tank that contains 120 kg of water at 21°C. Assuming the water that vaporizes during the process condenses back in the tank, determine the total entropy change during this process. **10**

b. Define and write the expression for below **04**

i. Thermal Conductivity

ii. Heat Flux

5. a. A household refrigerator that has a power input of 400 W and a COP of 1.8 is to cool 8 large muskmelons of 6 kg each to 8°C. If the muskmelos are initially at 32°C, determine how long it takes for the refrigerator to cool them. The muskmelons can be treated as water whose specific heat is 3.5 kJ/kg°C. What time period and power input would you suggest to cool it faster by 2 times. **07**

b. A heat engine receives heat from a heat source at 1200°C and has a thermal efficiency of 40%. The heat engine does maximum work equal to 500 kJ. Determine the heat supplied to the heat engine by the heat source, the heat rejected to the heat sink, and the temperature of the heat sink. **07**

OR

6. Consider heat is conducted in a rectangular tiffin box at a point in its coordinate system (x, y, z, t) at a length of "l", density of "ρ", and specific heat of "c". Derive the 1D heat conduction equation and express it for a constant thermal conductivity. **14**

7. a. The resistance wire of an 800W iron is 15 inch long and has a diameter of D = 0.08 inch. Determine the rate of heat generation in the wore per unit volume, in W/m^3, and the heat flux on the outer surface of the wire, in W/m^3, as a result of this heat generation. **07**

b. In a certain experiment, cylindrical samples of diameter 6 cm and length 11 cm are used. Two thermocouples in each sample are placed 3 cm apart. After initial transients, the electric heater is observed to draw 0.9A and 145V, and both differential thermometers read a temperature difference of 10°C. Determine the thermal conductivity of the sample. **07**

OR

8. a. An average man has a body surface area of 1.8 m^2 and a skin temperature of 33°C. The convection heat transfer coefficient for a clothed person walking in still air is expressed as $h = 8.6V^{0.53}$ for $0.5 < V < 2$ m/s, where V is the walking velocity in m/s. Assuming the average surface temperature of the clothed person to be 30°C, determine the rate of heat loss from an average man walking in still air at 10°C by convection at a walking velocity of a) 0.5 m/s, b) 1.0 m/s, c) 1.5 m/s and d) 2.0 m/s. **08**

b. During air cooling of potatoes, the heat transfer coefficient is combined convection, radiation and evaporation is determined experimentally to be: **06**

Air velocity (m/s)	Heat Transfer Coefficient ($W/m^{2o}C$)
0.66	14.0
1.00	19.1

Consider an 8 cm diameter potato initially at 20 °C with a thermal conductivity of 0.49 W/m°C . Potatoes are cooled by refrigerated air at 5°C at a velocity of 1m/s. Determine the initial rate of heat transfer from a potato and also to various velocities, and the initial value of the temperature gradient in the potato at the surface.